

POC System Link Budget Summary

Link Type	Distance	Elevation	Path Loss	LoS Prob	Status
HAPS ₁ → User	20.15 km	45.2°	142 dB	0.28	✓ Valid
HAPS ₂ → User	20.22 km	48.3°	142 dB	0.32	✓ Valid
HAPS ₃ → User	20.18 km	46.8°	142 dB	0.30	✓ Valid
UAV → User	0.3-5 km	10-85°	92-110 dB	Variable	✓ Relay
HAPS ₁ → Gateway	20.15 km	45.2°	Feeder	0.28	✓ Backhaul
HAPS ₂ → Gateway	20.22 km	48.3°	Feeder	0.32	✓ Backhaul
HAPS ₃ → Gateway	20.18 km	46.8°	Feeder	0.30	✓ Backhaul

Notes: All elevation angles > 30° regulatory minimum (ECC REPORT 156). Gateway elevation angles ensure beam tracking stability. LoS probabilities computed for urban environment ($\alpha=0.3$, $\beta=500$).